Miracle Seeds

AP Human Geography · Unit 5 · No stimulus (like Question 1) · 7 points · Teacher key · GeoArcade FRQ Tutor

No stimulus. Since 1950 the world's population has more than tripled to over 8 billion, and farmers now grow enough food to feed everyone, yet hundreds of millions of people still go hungry. This FRQ asks how farming changed to feed a growing world, who got left out, and how to keep it going without wearing out the land.

A. Identify ONE agricultural innovation associated with the Green Revolution. IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Names high-yield seed varieties (HYVs)
- Names another input: fertilizer, pesticides, irrigation, or machines

MODEL ANSWER

High-yield varieties (HYVs) of wheat and rice, like the short, sturdy wheat Norman Borlaug bred in Mexico.

Common mistake: “Farming got better.” That's a result, not an innovation. Name the actual new thing, like high-yield seeds or chemical fertilizer.

Why the model works: It names a specific innovation (high-yield varieties), the crops (wheat and rice), and a real example (Borlaug's wheat in Mexico).

B. Describe ONE way the Green Revolution changed food production in a specific country, such as India or Mexico. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a change: higher yields, more food, or self-sufficiency
- Gives proof from a real country (India, Mexico...)

MODEL ANSWER

T1- The Green Revolution greatly increased how much food India's farms produced per acre.
E2- Starting in the mid-1960s, farmers in the Punjab region planted high-yield wheat with chemical fertilizer and irrigation, and India's wheat harvest about doubled in less than ten years.
A3- This turned India from a country that depended on U.S. wheat shipments into one that, by the late 1970s, grew enough grain to feed its own people.

Common mistake: “The Green Revolution made farming better in India.” “Better” is too vague. Say WHAT changed (much bigger wheat and rice harvests) and give proof, like India's wheat harvest doubling.

Why the model works: It names the change (more food per acre), proves it with a real place (Punjab, India's harvest doubling), and describes the result (India could feed itself).

C. Explain ONE negative environmental consequence of the Green Revolution. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a harm: groundwater depletion, salty soil, pollution, or lost crop diversity
- Explains how Green Revolution farming causes it

MODEL ANSWER

T1- One negative environmental consequence of the Green Revolution is the depletion of groundwater.
E2- In India's Punjab region, high-yield rice and wheat need heavy irrigation, so farmers pump water from tube wells faster than rain can refill the aquifer, and the water table has dropped year after year.
A3- Because the farms depend on that water, a falling water table threatens to make the high yields impossible to keep up, putting future food production at risk.

Common mistake: "The Green Revolution was bad for the environment." That just repeats the question. Name a specific harm, like falling groundwater, and explain how the farming caused it.

Why the model works: It names a specific harm (groundwater depletion), uses a real place (Punjab), and explains the chain from thirsty crops to a falling water table.

D. Explain ONE reason the Green Revolution benefited large farms more than small farms. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a reason: the inputs were costly, and big farmers had more money or credit
- Explains how that helped large farms or hurt small ones

MODEL ANSWER

T1- The Green Revolution helped large farms more because its high-yield package was expensive, and wealthy farmers could afford it more easily.
E2- In India, the new seeds needed costly fertilizer, pesticides, and irrigation pumps to reach their highest yields, and large landowners could get bank loans using their land as collateral, while small farmers often borrowed from moneylenders at high interest.
A3- So large farms earned bigger profits and bought more land, while many small farmers fell into debt and some sold their land, widening the gap between rich and poor farmers.

Common mistake: "Large farms have more space to grow crops." True of any farming. It doesn't explain what about the Green Revolution helped them: the costly inputs big farmers could afford.

Why the model works: It names the reason (an expensive package), gives proof (bank loans vs. moneylenders), and explains the result (profits for big farms, debt for small ones).

E. Explain ONE reason the Green Revolution had less impact in sub-Saharan Africa than in South Asia. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a reason: different staples, little irrigation, high costs, or poor roads
- Explains how that limited the benefits

MODEL ANSWER

T1- The Green Revolution had less impact in sub-Saharan Africa because its high-yield seeds were bred for crops and conditions that didn't match most African farms.
E2- Scientists focused on irrigated wheat and rice, but many farmers in sub-Saharan Africa grow staples like cassava, sorghum, and millet on rain-fed fields, and only a small share of the region's farmland is irrigated.
A3- Since the new seeds didn't fit Africa's main crops or its dependence on rainfall, yields there rose much less than in South Asia.

Common mistake: "Africa is a desert, so nothing grows there." Not true: most of sub-Saharan Africa is savanna, farmland, and rainforest. Give a real reason, like seeds bred for irrigated wheat and rice instead of cassava and millet.

Why the model works: It names a clear reason (seeds bred for other crops), gives proof (cassava, sorghum, millet; little irrigation), and explains why yields rose less.

F. Explain ONE example of a sustainable farming practice and how it reduces environmental harm caused by agriculture. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a sustainable practice (no-till, crop rotation, terracing, organic, CSA...)
- Explains how it reduces a specific harm (erosion, water use, chemicals)

MODEL ANSWER

T1- One sustainable farming practice is conservation tillage, such as no-till farming.
E2- Instead of plowing the whole field each year, many farmers in places like the U.S. Midwest, Brazil, and Argentina plant seeds straight into the soil through the stalks and leaves left over from last year's crop.
A3- Because that leftover plant cover protects the ground from wind and rain, much less soil erodes away, which keeps fields fertile and keeps mud out of nearby rivers.

Common mistake: "Sustainable farming helps the environment because it is eco-friendly." That's circular: it just says sustainable is good for nature. Name ONE actual practice, like no-till or terracing, and explain HOW it reduces a specific harm.

Why the model works: It names a real practice (no-till), shows where it's used (the U.S. Midwest, Brazil, Argentina), and explains the cause → effect (plant cover → less erosion).

G. Explain what a food desert is and ONE reason food deserts exist in urban or rural areas of more developed countries. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Defines a food desert: little access to affordable, healthy, or fresh food
- Explains a reason: stores avoid poor or sparse areas, or residents lack cars

MODEL ANSWER

T1- A food desert is an area where people have little access to affordable, healthy food because there are few or no full-service grocery stores nearby.
E2- In many low-income city neighborhoods, such as parts of Detroit and Chicago's South Side, supermarket chains closed stores and opened new ones in the suburbs, where customers had more money to spend.
A3- Because grocery companies locate where they expect the highest profits, residents without cars often have to rely on corner stores and fast food, which makes a healthy diet harder and more expensive to get.

Common mistake: "A food desert is a place where there is no food, like the Sahara." That's the most common mix-up! A food desert usually has food (chips and soda at the corner store) but little affordable, healthy food. And you still need a reason it exists.

Why the model works: It defines the term (little access to affordable, healthy food), gives real places (Detroit, Chicago's South Side), and explains the cause (stores follow profits).

Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.